

Lecture 4: Calculus and Linear Algebra

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Course content (math tapas)

Sequences
and series

Functions
of one variable

Differentiation

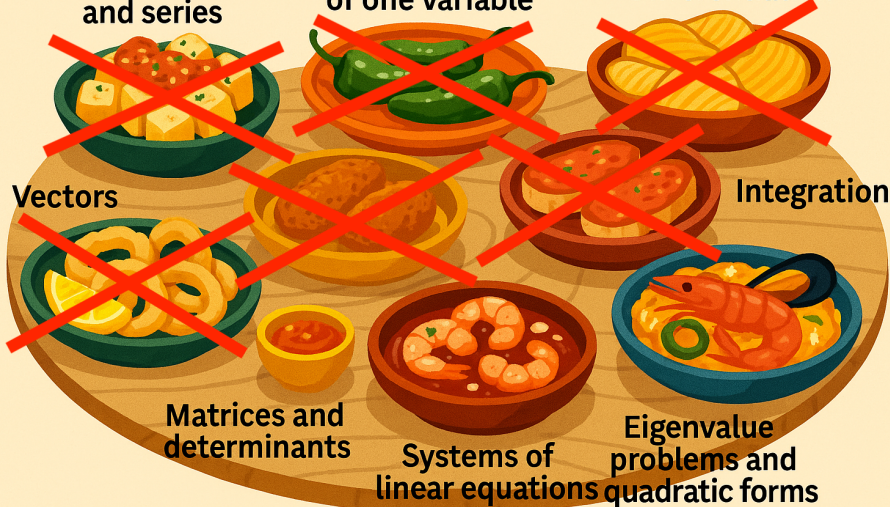
Vectors

Integration

Matrices and
determinants

Systems of
linear equations

Eigenvalue
problems and
quadratic forms



Linear subspaces

Linear subspace

The set $V \subset \mathbb{R}^n$ is a **linear subspace** of $\mathbb{R}^n$ if

$$\forall \mathbf{u}, \mathbf{v} \in V \quad \forall a, b \in \mathbb{R} \quad a\mathbf{u} + b\mathbf{v} \in V.$$

(closure under taking linear combinations)

There are two ways we can describe a linear subspace V :

- $V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ for some vectors $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$.
- Solution set of $A\mathbf{x} = \mathbf{0}_m$ for some $A \in \mathbb{R}^{m \times n}$.

Some examples:

- kernel of $A \in \mathbb{R}^{m \times n}$
- image of A
- Points in $\mathbb{R}^3$ satisfying $x_1 + x_2 + x_3 = 0$.

Why subspaces must have this form?

Let $V \subseteq \mathbb{R}^n$ be a linear subspace.

Take any nonzero $\mathbf{v}_1 \in V$. Then $\text{span}(\mathbf{v}_1) \subseteq V$.

If $\text{span}(\mathbf{v}_1) \subseteq V$, we are done.

If not, take any nonzero $\mathbf{v}_2 \in V \setminus \text{span}(\mathbf{v}_1)$.

Then $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent and again $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} \subseteq V$.

We proceed iteratively until we reach step m such that

$$V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}.$$

This procedure must stop as we cannot have more than n linearly independent vectors in $V \subseteq \mathbb{R}^n$.

System of linear equations

Systems of linear equations

Let A be an $m \times n$ matrix, $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ unknown, and $\mathbf{b} \in \mathbb{R}^m$ given. The system

$$A\mathbf{x} = \mathbf{b}$$

has the **vector form**

$$\sum_{j=1}^n x_j \mathbf{a}_j = \mathbf{b},$$

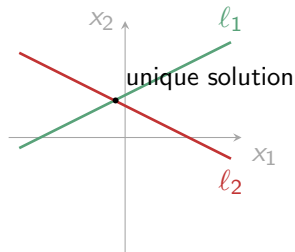
where $\mathbf{a}_1, \dots, \mathbf{a}_n$ are the columns of A .

If $\mathbf{b} = \mathbf{0}$ the system is **homogeneous**; otherwise it is **inhomogeneous**.

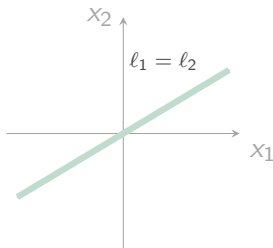
An inhomogeneous system can be **inconsistent** (no solution) or **consistent** (at least one solution).

A homogeneous system always has $\mathbf{x} = \mathbf{0}$ as a solution.

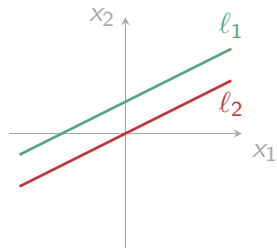
Geometry of 2×2 linear systems



Unique solution



Infinitely many solutions



No solution (inconsistent)

Key: Lines are the solution sets of single linear equations. Intersecting $\Rightarrow$ one point.
Coincident $\Rightarrow$ all points. Parallel $\Rightarrow$ none.

Augmented matrix; consistency

The **augmented matrix** of the linear system $A\mathbf{x} = \mathbf{b}$ is

$$A_{\mathbf{b}} = [A \ \mathbf{b}] \in \mathbb{R}^{m \times (n+1)}.$$

Remark: Either $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$ or $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A) + 1$.

Theorem (Rouché–Capelli)

$A\mathbf{x} = \mathbf{b}$ is **consistent** $\iff \text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$ (equivalently $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$).

Moreover:

1. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = n$, the solution is **unique**. **Q:** What is it?
2. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) < n$, there are **infinitely many** solutions.

The rank tells us how many variables are **pivots** (determined by equations).

The remaining variables can be chosen freely, hence called **free variables**.

Solving with Gaussian elimination

Row echelon forms

A matrix is in **row echelon form (REF)** if:

1. All nonzero rows appear above any zero rows.
2. Each **pivot** (leading entry in a row) is to the right of the pivot in the row above it.
3. All entries below a pivot are zero.

It is in **reduced row echelon form (RREF)** if, in addition:

1. Each pivot is equal to 1.
2. Each pivot is the only nonzero entry in its column.

Example:

$$\begin{pmatrix} \mathbf{1} & 2 & 0 & 3 \\ 0 & \mathbf{1} & 4 & -1 \\ 0 & 0 & 0 & \mathbf{1} \end{pmatrix} \quad (\text{REF, not RREF}). \qquad \begin{pmatrix} \mathbf{1} & 0 & 0 & 2 \\ 0 & \mathbf{1} & 0 & -3 \\ 0 & 0 & \mathbf{1} & 5 \end{pmatrix} \quad (\text{RREF}).$$

Gaussian elimination and RREF

Theorem

By applying elementary row operations to the augmented matrix $A_{\mathbf{b}}$, any system $A\mathbf{x} = \mathbf{b}$ can be transformed into an equivalent system in **reduced row echelon form (RREF)**.

$$\begin{array}{rcl} x_1 + x_2 + x_3 & = & 3 \\ x_1 - x_2 + 2x_3 & = & 2 \\ 4x_1 + 6x_2 - x_3 & = & 9 \end{array} \iff \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

So the unique solution is $\mathbf{x} = (1, 1, 1)$.

Why equivalence holds:

$A\mathbf{x} = \mathbf{b} \iff A_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0}$. Row operations correspond to multiplying $A_{\mathbf{b}}$ on the left by an invertible matrix C , so the solution set is unchanged.

A homogeneous example

Consider

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 0,$$

$$2x_1 + 2x_2 + 2x_3 + 2x_4 - x_5 - x_6 = 0,$$

$$3x_1 + 3x_2 - x_3 - x_4 - x_5 - x_6 = 0.$$

Row-reducing the coefficient matrix gives

$$\begin{pmatrix} \color{red}{1} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \color{red}{1} & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \color{red}{1} & 1 \end{pmatrix}.$$

So there are 3 pivots, $\text{rank}(A) = 3$, and the solution space has dimension $6 - \text{rank}(A) = 3$:

$$(x_1, -x_1, x_3, -x_3, x_5, -x_5).$$

A basis is $\{(1, -1, 0, 0, 0, 0), (0, 0, 1, -1, 0, 0), (0, 0, 0, 0, 1, -1)\}$.

General solution structure

Theorem

Every solution of $A\mathbf{x} = \mathbf{b}$ can be written as $\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p$, where $\mathbf{x}_h$ is the general solution of the homogeneous system $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x}_p$ is any particular solution of $A\mathbf{x} = \mathbf{b}$.

Corollary: If $A\mathbf{x} = \mathbf{b}$ is consistent, then the set of solutions is an affine subspace parallel to $\ker(A)$.

Example: Three identical equations

$$x_1 + x_2 + x_3 = 1$$

reduce to

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \rightsquigarrow \begin{array}{l} x_1 = 1 - x_2 - x_3, \\ x_2, x_3 \text{ free.} \end{array}$$

So

$$\mathbf{x} = (1, 0, 0) + \lambda_1(-1, 1, 0) + \lambda_2(-1, 0, 1).$$

Recipe: solving $A\mathbf{x} = \mathbf{b}$ with Gaussian elimination

1. Write down the augmented matrix $[A \mid \mathbf{b}]$.
2. Use row operations to reduce to RREF.
3. Identify pivot and free variables.
4. Express the solution: $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$.

Always

- The number of pivot variables is the rank of A ,
- the number of free variables $= n - \text{rank}(A)$,
- the free variables parametrize the solution set.

Two quick examples

1. For $A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix}$, RREF gives $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$. So $\text{rank}(A) = 2$, $\dim \ker(A) = 1$, and

$$\ker(A) = \{\lambda(-2, 1, 1) : \lambda \in \mathbb{R}\}.$$

2. For $\begin{cases} x_1 + x_2 + x_3 = -1 \\ x_1 + 2x_2 + 4x_3 = 2 \end{cases}$, the augmented matrix reduces to $\left(\begin{array}{ccc|c} 1 & 0 & -2 & -4 \\ 0 & 1 & 3 & 3 \end{array} \right)$. So $\text{rank}(A) = 2$, one free variable $x_3 = \lambda$, and

$$\mathbf{x} = (-4, 3, 0) + \lambda(2, -3, 1).$$

Finding a basis for a span

Let

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}.$$

Put them as columns of a matrix:

$$M = \begin{pmatrix} 2 & -1 & 0 \\ 3 & 0 & -1 \\ 0 & 3 & -2 \end{pmatrix}.$$

Row-reduction gives $M \sim \begin{pmatrix} 1 & 0 & -1/3 \\ 0 & 1 & -2/3 \\ 0 & 0 & 0 \end{pmatrix}.$

Thus $\text{rank}(M) = 2$ and there is one linear dependence: $3\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$.
A basis for $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ can be taken as the first two pivot columns, e.g.

$$\{\mathbf{v}_1, \mathbf{v}_2\}.$$

Quick recap

- **Homogeneous system:** $Ax = 0$. Always has at least the trivial solution.
- **Inhomogeneous system:** $Ax = b$. May or may not be solvable.
- **Augmented matrix:** $[A \mid b]$, i.e. stick b as a new column.
- **Rank:** Number of pivot columns after row reduction = dimension of the column space.

Standing on the shoulders of giants



“If I have seen further, it is by standing on the shoulders of giants.” **Isaac Newton (1675).**

Some of the giants of linear/matrix algebra:

- Gauss (Gaussian elimination, least squares, determinants)
- Cauchy (Cauchy–Schwarz, eigenvalues, determinants)
- Laplace (Laplace expansion of determinants)
- Cayley (Cayley–Hamilton theorem, matrix theory)
- Sylvester (matrix rank, invariant theory)
- Jordan (Jordan canonical form)
- Frobenius (matrix factorizations, linear algebra foundations)

(pictured here as the base of a *castell*)

Chapter 8: Eigenvalues and Quadratic Forms

Eigenvalues/eigenvectors appear in growth models, Markov chains, PCA, and stability of dynamical systems.

Quadratic forms drive optimization via second-order conditions.

Read Chapter 10 of Werner–Sotskov (optional: Simon–Blume, Ch. 7).

Exercises 10.2; 10.5 A,B,C; 10.6(a,b).

Eigenvalues and eigenvectors

Eigenvalues and eigenvectors: definitions

For $A \in \mathbb{R}^{n \times n}$, a scalar λ is an **eigenvalue** if there exists $\mathbf{x} \neq \mathbf{0}$ with

$$A\mathbf{x} = \lambda\mathbf{x} \iff (A - \lambda I_n)\mathbf{x} = \mathbf{0}.$$

Then $\mathbf{x}$ is an **eigenvector** for λ .

Characterization:

$$\lambda \text{ eigenvalue} \iff \text{rank}(A - \lambda I_n) < n \iff \det(A - \lambda I_n) = 0.$$

The **characteristic polynomial** $P(\lambda) = \det(A - \lambda I_n)$ has degree n ; its real roots (at most n) are the real eigenvalues.

Procedure:

1. Solve $\det(A - \lambda I_n) = 0$ for eigenvalues.
2. For each λ , solve $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ for eigenvectors.

e.g. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

Why eigenvalues matter (at a glance)

If $A\mathbf{x} = \lambda\mathbf{x}$ then $A^k\mathbf{x} = A^{k-1}(A\mathbf{x}) = \lambda A^{k-1}\mathbf{x} = \dots = \lambda^k\mathbf{x}$.

- **Markov chains:** steady state = eigenvector for $\lambda = 1$.
- **Growth/dynamics:** long-run rate $\approx$ spectral radius $\rho(A)$.
- **PCA:** principal directions = eigenvectors of covariance; explained variances = eigenvalues.
- **Quadratic forms:** curvature of $\mathbf{x}^\top A\mathbf{x}$ aligned with eigenvectors; definiteness from eigenvalue signs.

Eigenvalues and eigenvectors: migration example

Two regions: city (x_t^C) and countryside (x_t^R). Let $\mathbf{x}_t = (x_t^C, x_t^R)^\top$. Assume next-year proportions follow a **column-stochastic** transition:

$$\mathbf{x}_{t+1} = A \mathbf{x}_t, \quad A = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix},$$

so each column sums to 1 (fractions of people who *arrive* from each region).

A **stable composition** $\mathbf{u}$ satisfies $A\mathbf{u} = \mathbf{u}$ and $u_1 + u_2 = 1$:

$$(A - I)\mathbf{u} = 0 \Rightarrow \mathbf{u} = \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}.$$

Spectrum: $\lambda_1 = 1$, $\lambda_2 = 0.4$.

Any initial $\mathbf{x}_0$ with positive entries converges to $\mathbf{u}$.

- Check $\mathbf{v} = (1, -1)$ is the 0.4-eigenvector.
- $\mathbf{x}_0$ can be written as $\mathbf{x}_0 = \mathbf{u} + \lambda \mathbf{v}$
- $\mathbf{x}_t = A^t(\mathbf{u} + \lambda \mathbf{v}) = \mathbf{u} + \lambda \cdot 0.4^t \cdot \mathbf{v}$.
- $\|\mathbf{x}_t - \mathbf{u}\| = |\lambda| \cdot 0.4^t \sqrt{2} \rightarrow 0$.

Basic spectral facts

For $A \in \mathbb{R}^{n \times n}$

1. Eigenvectors associated with distinct eigenvalues are **linearly independent**.
2. If A is **symmetric**, then all eigenvalues are real and eigenvectors for distinct eigenvalues are **orthogonal** (Spectral Theorem).

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Then $P(\lambda) = \lambda^2 - 2\lambda - 3$, so $\lambda_1 = 3$, $\lambda_2 = -1$ with eigenvectors along $(1, 1)$ and $(-1, 1)$, respectively.

Matrix diagonalization

Diagonalizable or not?

Definitions:

- $A \in \mathbb{R}^{n \times n}$ is **diagonalizable** if there exists an invertible P such that

$$P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Equivalently, A has a basis of eigenvectors.

- For each eigenvalue λ , the **eigenspace** is

$$E_\lambda = \ker(A - \lambda I).$$

Its dimension is the **geometric multiplicity**.

- The number of times λ appears as a root of the characteristic polynomial is its **algebraic multiplicity**.

Why it matters: diagonalization simplifies A^k , e^A .

Fact:

A is diagonalizable $\iff \sum_{\lambda} \dim E_{\lambda} = n$.

Example 1 (diagonalizable).

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Characteristic polynomial: $(2 - \lambda)^2 - 1$. Distinct eigenvalues $\Rightarrow$ eigenvectors span $\mathbb{R}^2$. So A is diagonalizable: $A = PDP^{-1}$ with $D = \text{diag}(3, 1)$ and

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Example 2 (defective).

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Characteristic polynomial: $(\lambda - 1)^2$ (algebraic multiplicity 2). Eigenspace: $\ker(B - I) = \text{span}\{(1, 0)\}$ has dimension 1. So B is **not** diagonalizable.

Repeated eigenvalues

Consider

$$A = \begin{pmatrix} 0 & -1 & 1 \\ -7 & 0 & 5 \\ -5 & -2 & 5 \end{pmatrix}.$$

The characteristic polynomial is

$$p_A(\lambda) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 2)^2(\lambda - 1).$$

Hence the eigenvalues are $\lambda = 2$ (algebraic mult. 2) and $\lambda = 1$ (algebraic mult. 1).

Eigenspaces:

$$E_2 = \ker(A - 2I) = \text{span}\{(1, -1, 1)\}, \quad E_1 = \ker(A - I) = \text{span}\{(2, 1, 3)\}.$$

Since $\dim E_2 = 1 < 2$, A is defective.